CHIRAYU MAHAJAN

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AI Engineer focused on building end-to-end AI systems across computer vision, LLM agents, real-time inference, and backend infrastructure. Built systems spanning real-time UPI fraud detection, multi-agent marine intelligence, CCTV incident detection, and persistent LLM memory. Pursuing a B.Tech in Artificial Intelligence at MIT Academy of Engineering alongside a B.S. in Data Science from IIT Madras. Experienced in taking systems from architecture and implementation through deployment.

EXPERIENCE

AIDeveloperIntern — Tronocity Labs May 2026 – Present

Building Gigzs–Akhrot, a persistent memory & context layer for multi-LLM agent systems

- Built LLM pipelines and backend services that let a multi-agent memory product retain state across models, sessions, and agent handoffs instead of starting stateless every call.
- Co-designed a scalable architecture for cross-model reasoning; contributed to internal benchmarking that cut token usage by up to 58% in targeted scenarios.
- Worked full-stack across the product: LLM orchestration, retrieval/memory services, and backend APIs.

Product Development Intern — MITAOEEntrepreneurialDevelopmentFoundation(EDF) May – Jul 2026

TrafficBuddy — traffic complaint management app

- Drove feature planning, requirement analysis, UI/UX design, and application workflows; coordinated with the dev team on backend integration across multiple rounds of user feedback.

SELECTED PROJECTS

VISTA — Real-Time CCTV Accident & Incident Detection Github: [Link]

- Built a computer-vision pipeline (YOLO11m + ByteTrack + homography-based speed estimation) that detects road accidents from live CCTV in ~0.5–1s and auto-routes alerts to traffic police, EMS, or control room by severity.
- Designed 4 independent heuristic signals (speed-drop, collision, anomaly-stop, hit-and-run) with cross-frame verification to suppress false positives from traffic-light queues, validated against 10 real clips + synthetic ground-truth speed tests (±15% accuracy gate).

Iron Wallet—Real-Time UPI Fraud Prevention Engine Winner, Nirman 2026 (GDG) Github: [Link]

- Led the risk-scoring engine for a real-time payment fraud system, combining OTP verification with a behavioral risk model served over FastAPI + WebSockets.

Synapse — AI-Native Project Workspace Deployed: [Link]

- Shipped a full-stack (Next.js 15, Supabase, Groq) project tracker built for AI-assisted dev teams: kanban board, GitHub webhook auto-close, and an AI-powered codebase-analysis feature.
- Built a 'Context Bridge' tool that converts a Claude conversation transcript into a continuity brief, letting developers resume work in a fresh session without re-explaining context.

PrepHire.AI — AI-Powered Mock Interview & Placement Platform Deployed: [Link]

- Built a full-stack platform running resume-tailored, proctored AI voice interviews (Technical, HR, Aptitude, GD), scoring candidates on a 5-axis rubric with a concrete action plan, plus an integrated coding arena with AI code review and role-scoped portals for students, faculty, and admins across 9 engineering branches.

ORCA —AgenticMarineIntelligencePlatform Smart India Hackathon 2026 (ISRO, PS-26176)

- Co-built a multi-agent conversational platform (FastAPI + LangGraph) answering natural-language marine queries — fishing zones, sea safety, hazards — by orchestrating specialist agents over live INCOIS satellite/ocean data, with multilingual (11 Indic languages) and explainable, evidence-backed responses.

EDUCATION

B.Tech,Artificial Intelligence & Machine Learning — MIT Academy of Engineering Sep 2025 – Present

B.S.,Data Science and Applications — Indian Institute of Technology, Madras Sep 2025 – Present

SKILLS

Programming: Python, C, JavaScript/TypeScript, SQL
AI / ML: PyTorch, scikit-learn, XGBoost, OpenCV, Computer Vision, NLP, LLMs, RAG, Embeddings, Prompt Engineering
AI Systems & Agents: LangGraph, LangChain, Multi-Agent Systems, LLM Orchestration, Tool Calling, Agent Memory, Context Engineering, Evaluation
Backend & APIs: FastAPI, Flask, REST APIs, WebSockets, Async Python, Authentication, Webhooks
Data & Storage: PostgreSQL, MySQL, SQLite, Supabase, Firebase, Vector Databases, Redis
Frontend: React, Next.js, Tailwind CSS
Cloud & Infrastructure: AWS, Docker, Vercel, Render, Git/GitHub, CI/CD
Engineering: System Design, API Design, Database Design, Real-Time Systems, Distributed Systems, Software Architecture

CERTIFICATIONS & ACHIEVEMENTS

- Winner — Nirman 2026 (GDG) for PayTrust | Participant — TechSprint 2026 (Cyberfury) & Datathon 2026 (Team Hellfire)
- Cisco (Modern AI, Data Analytics, Python Essentials 1 & 2) | IBM SkillsBuild (Career Mgmt Essentials) | Anthropic (Claude 101)